

Check RMA Device Tool Guide

Revision History

Date	Version	Author	Note
2026-07-31	1.0	Hau Pham	Initial version

Overview

Hardware team receives RMA devices returned from the US, performs hardware debugging, and pulls each device's last log from its SD card. Those logs are handed to the firmware team, which runs this tool to check the logs against the stated RMA reason and decide whether the observed root cause matches it.

Requirements

- **Python 3+** — check:

```
python --version
```

- **SSH access to the customer host** — check:

```
ssh customer "echo ok"
```

- Run **setup.bat** once (creates **.venv**, installs the pinned deps):

```
setup.bat
```

Input

config.json points at two things — put them anywhere on disk:

Field	Meaning
device_logs_dir	folder with one subfolder per device (the raw logs)
output_dir	where report_rma.xlsx + charts are written

Device IDs are read from the logs automatically and used to fetch the RMA report — no device list to maintain. Paths may be absolute, or relative to **config.json**.

Example

Point **config.json** at the bundled 4-device sample — copy this in:

```
{
  "device_logs_dir": "examples/Initial Log",
  "output_dir": "examples/my_output"
}
```

Then run the whole pipeline with one command:

```
run.bat
```

Output — report_rma.xlsx

Tabs below are the report's sheets (from a sample run).

For Hardware team

The RMA report in its **original nested format** (same 3-column layout — Device ID / Device Information 1 / Device Information 2 — as the source RMA report CSV), copied in verbatim. The landing sheet.

For Hardware team (nested RMA report format, same as the source CSV)

Device ID	Device Information 1	Device Information 2
1325061309	(1) - Deploy date: 2025-10-11 (2) - RMA feedback date: 2026-05-04 (3) - Number of studies run: 24 (4) - Total number of study days: 88 days 12 minutes	(1) - LOT Devices: L20250905 (WO-250520) (2) - Reason for RMA: DAMAGED DESCRIBE (3) - SE findings: ECG LEADS DAMAGE CONFIRMED; BACK CASE DAMAGE; Other: carrying case damaged; Resolution: ECG CABLE REPLACED; BACK CASE REPLACED; Other: carrying case replaced
1325062494	(1) - Deploy date: 2026-01-09 (2) - RMA feedback date: 2026-04-14 (3) - Number of studies run: 14 (4) - Total number of study days: 22 days 16 hours 36 minutes	(1) - LOT Devices: L20251222 (WO-250520) (2) - Reason for RMA: CHARGING ISSUE (3) - SE findings: Other: devices sent to VN for testing, do not charge for RMA; Resolution: NON REPAIRABLE
1325060142	(1) - Deploy date: 2025-09-05 (2) - RMA feedback date: 2026-04-27 (3) - Number of studies run: 0 (4) - Total number of study days: 0 minutes	(1) - LOT Devices: L20250815 (WO-250520) (2) - Reason for RMA: RETURN TO STOCK (3) - SE findings: Other: devices sent to VN for testing, do not charge for RMA; Resolution: NON REPAIRABLE; Other: No issue found.

RMA_Comparison

Per device: RMA reason vs. log evidence, an **Explained by log?** verdict, and a date match.

RMA_Comparison (RMA reason vs. log evidence + verdict)

Device	Device ID	RMA reason - disposition	Category	Explained by log?	Explanation	Date match
32	1325061309	DAMAGED DESCRIBE - ECG LEADS DAMA...	PHYSICAL	N/A (mechanical)	physical/mechanical: N/A - not ob...	logs after RMA
34	1325062494	CHARGING ISSUE - Other: devices s...	CHARGING	Yes	charging: CORROBORATED (851 charg...	logs after RMA
39	1325060142	RETURN TO STOCK - Other: devices ...	CHARGING, RTS	Yes	charging: NOT clearly in logs (0 ...	logs after RMA
42	1325061793	DEVICE ERRORS DESCRIBE - Other: d...	INTERNAL_ERROR, CHARGING	Yes	internal-error: CORROBORATED (1 f...	in window

Summary

One row per device: FW/HW, boots by reason, boot-issue / DETECTION counts, faults.

- **Error** — number of reboots caused by a firmware fault (a boot preceded by an **Error report** - **Begin** block). This differs from **Err reports** = the total number of fault events logged (including **AFE Recovery failed**); not every fault event causes a reboot.
- **Boot issue (no charge)** — see Power_Loss below.

Summary (one row per device)

Device	PROV	Device ID	FW ver	HW ver	Boots	Normal	Error	Boot issue (no charge)	DETECTION	All faults (name x count)
32	PROV2	1325061309	1.0.0.40a	2.1	4	4	0	0	0	
34	PROV2	1325062494	1.0.0.40	2.1	4	3	0	1	851	
39	PROV2	1325060142	1.0.0.40	2.1	3	2	0	0	0	
42	PROV2	1325061793	1.0.0.40	2.1	5	5	0	2	2	BSP_ERR_TI_AFE_CHIP_RECOVERY×1

Power_Loss

How the device went down before each boot; the **Boot issue (no charge)** flag is the hard evidence for a charging defect.

Power_Loss (how the device went down before each boot)

Device	Seq	Timestamp	Power event	Boot source	Boot issue (no charge)
32	1	2026-06-03 00:17:46	clean	AppRequest	
32	2	2026-06-03 16:11:58	clean	Charger	
32	3	2026-06-03 16:18:14	clean	AppRequest	
32	4	2026-07-07 08:26:26	clean	PwrKey	
34	1	2026-05-06 15:22:50	power_on	Charger	yes
34	2	2026-05-06 20:36:42	clean	AppRequest	
34	3	2026-05-07 20:28:34	clean	AppRequest	
34	4	2026-07-07 08:45:51	clean	PwrKey	
39	1	2026-05-19 15:28:50	power_on	Charger	
39	2	2026-05-19 17:38:13	clean	AppRequest	
39	3	2026-07-07 09:38:47	clean	PwrKey	
42	1	2026-04-28 15:15:44	clean	AppRequest	

What Boot issue (no charge) means. It flags a boot from **Boot source: Charger** (the device was off and powered on when the charger was plugged in) where charging is not healthy — either it never reaches a charging state (**FAST_CC** / **FAST_CV** / **COMPLETED** / **TRICKLE**, staying **OFF** / **DETECTION** / **SUSPEND**), or it drops out of charging back to **DETECTION** / **OFF** **twice or more** (flapping contact). Real example from device 34 (a "CHARGING ISSUE" RMA) — the charge state keeps flapping within one charger session:

```
[260506:152245] Boot source: Charger      <- powered on by the charger
260506:152252   charge stat=FAST_CC
260506:181216   charge stat=DETECTION      <- drop out of charging
260506:181219   charge stat=FAST_CC
260506:181220   charge stat=OFF           <- drop out of charging
260506:202003   charge stat=DETECTION      <- drop out of charging
260506:203525   charge stat=DETECTION      <- drop out of charging
...             (8 drops in this one charger session -> boot issue = yes)
```

A healthy charge is one initial **DETECTION**, then sustained **FAST_CC/FAST_CV** with at most one final drop (unplug) — so it never reaches the ≥ 2 threshold.

Fault_Counts

BSP fault codes per device, with priority + firmware action.

Fault_Counts (BSP faults per device)

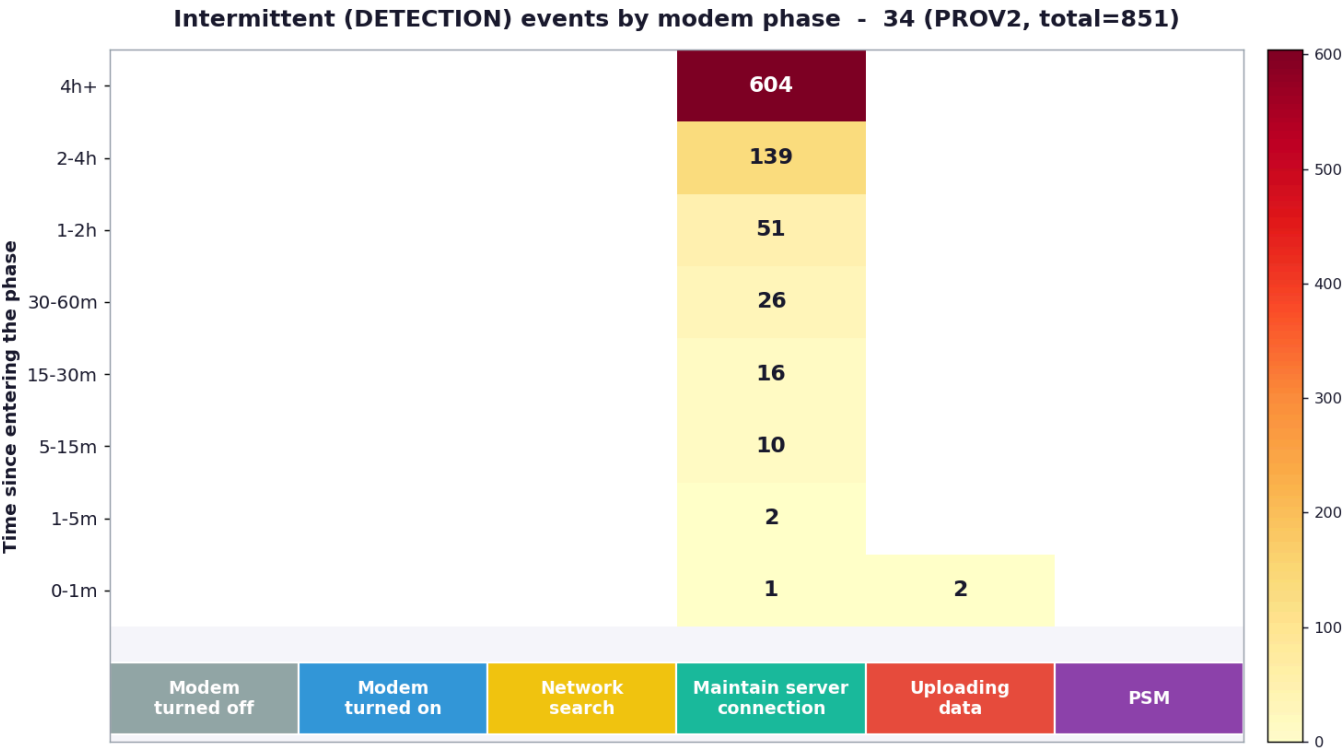
Device	BSP	Error name	Priority	Action	Count
42	20402	BSP_ERR_TI_AFE_CHIP_RECOVERY	2	Emergency stop	1

Modem_Detection + Charts

Time & **charge stat=DETECTION** count per modem phase, plus a per-device heatmap of when the intermittent detections happen (modem phase × time-into-phase).

Modem_Detection (time & DETECTION per modem phase)

Device	Phase	Duration (h)	DETECTION count	DET / min
32	no_LTE_modem	0.08	0	0
32	turn_on_LTE_modem	15.92	0	0
32	searching	0.03	0	0
32	non_uploading	808.43	0	0
32	uploading	0	0	0
32	psm	0	0	0
32	psm_ble	0	0	0
34	no_LTE_modem	0.5	0	0
34	turn_on_LTE_modem	1433.41	0	0
34	searching	0.03	0	0
34	non_uploading	27.13	849	0.522
34	uploading	3.33	2	0.01
34	psm	17	0	0
34	psm_ble	0	0	0



Other sheets: [Fault_Totals](#) (fleet-wide fault tallies), [Error_Reports](#) (raw parsed error blocks), [SD_Corruption](#) / [Corrupt_Logfiles](#) (storage corruption), and the fleet health [Charts](#) overview.

Reading the verdict ([Explained by log?](#))

- **Yes** — logs corroborate the RMA reason (e.g. charging RMA + boot-issue / DETECTION; device-errors RMA + matching BSP faults).
- **No** — RMA claims a fault the logs do not show (worth a closer look).
- **N/A (mechanical)** — physical damage (lead/case) is not observable in logs; the logs only confirm the device was still electrically active.
- **Partial / ?** — mixed or insufficient evidence.

A device can read "No fault in logs" simply because the relevant log was overwritten by garbage — check [Corrupt_Logfiles](#) too.

Notes

- No customer/patient data is committed beyond the reviewed [examples/](#) sample (see [../CONTRIBUTING.md](#) §8).
- [input-tool/run-remote-503.sh](#) is a backend-owned script, vendored here — do not modify it in this repo.
- Individual stages can still be run by hand ([src/append_cus503_to_report.py](#), [src/make_input_csv.py](#), [src/analyze_rma.py](#)) — see [CLAUDE.md](#).